

Smart Metro Ticket Booking System

Project Design Phase

Date

Team ID

Project Name: Smart Metro Ticket Booking System

Maximum Marks: 4 Marks

3. Problem-Solution Fit

This document lines up the pain points each persona faces under the current manual, counter-based metro ticketing process against the capabilities the proposed ServiceNow solution provides to fix them.

3.1 Problem Statement

Today, buying a metro ticket depends heavily on physical counters and handling cash. Passengers deal with long queues during peak commute hours, limited digital payment acceptance, and no way to look back at past bookings. Station Managers update fare and station information by hand and validate tickets without a real-time, auditable check. Metro Operations has no consolidated view of ticket volume, revenue split, or peak-time demand, leaving schedule and resource planning reactive instead of data-driven. IT Admins can't change fare rules or catalog configuration without filing a development request, which slows down routine business changes.

Figure 5.1 — Mapping of persona-level pain points to the proposed solution fit

3.2 Problem — Solution Fit Mapping

Persona	Problem / Pain Point	Solution Fit
Passengers	Long queues, cash-only counters, no digital ticket or booking history.	Self-service Service Catalog booking with instant QR ticket delivery and a viewable booking history.
Station Managers	Manual station/fare updates and slow, unverifiable ticket checks at the gate.	Controlled, approval-based master-data workflow plus real-time QR validation within 2 seconds.
Metro Operations Team	No consolidated view of ticket volume, revenue, or peak travel patterns.	Performance Analytics dashboards surfacing usage, digital-vs-cash revenue split, and peak-time trends.
IT Admins	Fare and catalog changes require code deployment, slowing routine updates.	No-code configuration of catalog variables, fare rules, and ACLs directly in the ServiceNow UI.

3.3 Why Now (Market & Timing Fit)

- Rising smartphone and UPI/digital-wallet adoption means most passengers already have what they need to pay and receive tickets digitally.
- Post-pandemic commuter expectations favour contactless, counter-free ticketing over handling cash.
- The metro network is already planning QR-based Automated Fare Collection (AFC) gate hardware, making this the natural point to digitize the booking front end.